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324613 (24)

BE (6th Semester)

Examination, April-May, 2014

Branch : Electrical, EEE

POWER ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : Attempt any two parts from (b), (c) and (d) of each question. Part (a) of each question is compulsory.

Q. 1. (a) Explain, why dv/dt must be limited ? 2

(b) Discuss what would happen if gate is made positive with respect to cathod during reverse blocking of a thyristor.

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Define latching and holding current as applicable to an SCR. Show these current on its static V–I characteristics. 7

(c) Explain how the voltage equalization takes place when SCR's are connected in series, and what requirements should be satisfied by a gate trigger. 7

(d) Draw the circuit of a relaxation oscillator using UJT. Show output voltage wave-shape 7

Q. 2. (a) What is difference between source and natural commutation ? 2

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(b) Draw input & output waveform of single phase full-wave controller with R-L load Explain working of the circuit. 7

(c) A single phase fully-controlled bridge converter supplies an inductive load. Assuming that output current is virtually constant, and is equal to I_d, determine following performance measure, if the supply voltage is 230 V and firing angle is maintained at $\pi/6$: 7

(i) Average output voltage

(ii) Supply RMS current

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(iii) Supply fundamental RMS current

(iv) Fundamental power factor

(v) Supply power factor

(vi) Supply harmonic factor

(vii) Supply ripple factor.

(d) How power factor improve in phase control rectifier ? 7

Q. 3. (a) What is the purpose of the commutation circuit of a chopp ? 2

(b) Explain in brief chopper classification. 7

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(c) Describe a morgan chopper with associate voltage and current waveform. Enumerate the demerit of morgan chopper compare to jones chopper. 7

(d) For the ideal type A chopper circuit, following conditions are given : 7

E_{dc} = 220 V, chopping frequency = 500 Hz, duty cycle α = 0.3, R = 1 Ω, L = 3 mH and E_b = 23. Compute the following :

(i) Check whether the load current is continuous or not.

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(ii) Average output current.

(iii) Maximum and minimum value steady state output current.

(iv) RMS value of first, second and thir harmonics of load current.

(v) Average value of source current.

(vi) Input power, power absorbed by back emf E_b and power loss in resistor.

(vii) RMS value of output current.

Q. 4. (a) What is delay in inverter circuit. 2

(b) Explain in brief modified series inverter. 7

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(c) A single phase-bridge inverter delivers power to a series circuit connected RLC load with R = 3 Ω, X_L = 12 Ω. The periodic time T = 0.2 ms. What value of C should the load have in order to obtained load commutation for the SCRs. The thyristor turn-off time is 12 μs. Assume that the load current contains only fundamental component. 7

(d) Explain with the help of circuit diagram and waveforms single phase to single-phase cyclo convertor. 7

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Q. 5. (a) What is the purpose of a converter in D derive ? 2

(b) Explain four quadrant chopper drives. 7

(c) Write short notes on : 7

(i) MOSFET

(ii) Modern semiconductor device.

(d) Explain following : 7

(i) JGBT

(ii) Power transistor.

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